

CLASSIFICATION MODELS, EVALUATION METRICS AND HANDLING IMBALANCED DATA

Breast Cancer Classification Using Machine Learning

ABSTRACT

This project focuses on developing and evaluating classification models using the Breast Cancer Wisconsin Dataset. Logistic Regression and Decision Tree classifiers were implemented to classify tumors as malignant or benign. The workflow included data preprocessing, feature scaling, model training, prediction, and evaluation. Performance was assessed using Confusion Matrix, Precision, Recall, F1-Score, ROC Curve, and AUC Score. Class imbalance handling was also explored using balanced class weights. The results demonstrated that Logistic Regression provides reliable and accurate classification performance for breast cancer prediction.

1. INTRODUCTION

Machine Learning enables computers to learn patterns from data and make predictions without being explicitly programmed. Classification is a supervised learning technique used to predict categorical outcomes and is widely applied in healthcare, fraud detection, and recommendation systems.

In this project, a binary classification model was developed using the Breast Cancer Wisconsin Dataset. The objective was to classify tumors as malignant or benign and evaluate the effectiveness of different classification techniques using multiple performance metrics.

2. DATASET ANALYSIS

The Breast Cancer Wisconsin Dataset was obtained from the Scikit-learn library. It contains diagnostic measurements computed from digitized images of breast mass samples.

Dataset Characteristics

- Total Samples: 569
- Features: 30 numerical attributes
- Classes: 2

Target Labels:

- 0 – Malignant
- 1 – Benign

Observations

- The dataset contains no missing values.
- All features are numerical and suitable for machine learning models.
- Feature scaling is required for improved model performance.
- The dataset represents a binary classification problem.

3. MODEL BUILDING

The dataset was divided into input features (X) and target labels (y). The data was split into training and testing sets using an 80:20 ratio with stratified sampling to preserve class distribution.

Feature scaling was performed using Standard Scaler to normalize the data before training.

Two classification algorithms were implemented:

Logistic Regression

Logistic Regression was trained on the scaled dataset and used as the primary classification model.

Decision Tree Classifier

A Decision Tree model was trained and evaluated to compare its performance with Logistic Regression.

The implementation was carried out using Python libraries including Pandas, NumPy, Matplotlib, and Scikit-learn.

4. EVALUATION METRICS

The performance of the models was evaluated using several classification metrics.

Confusion Matrix : The Confusion Matrix provides information about:

- True Positives (TP)
- True Negatives (TN)
- False Positives (FP)
- False Negatives (FN)

It helps identify the number of correct and incorrect predictions made by the classifier.

Precision: Precision measures the proportion of correctly predicted positive observations among all predicted positive observations.

Recall: Recall measures the proportion of actual positive observations correctly identified by the model.

F1-Score: F1-Score combines Precision and Recall into a single metric and is useful when both false positives and false negatives are important.

ROC Curve and AUC Score: The ROC Curve illustrates the relationship between True Positive Rate and False Positive Rate. The AUC Score measures the model’s ability to distinguish between classes, with higher values indicating better performance.

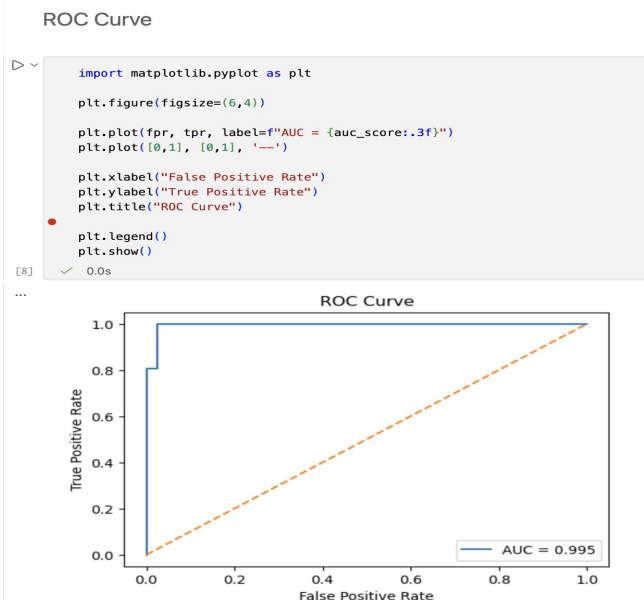


Figure 1: ROC Curve and AUC Score

```
tree = DecisionTreeClassifier(random_state=42)

tree.fit(X_train, y_train)

y_pred_tree = tree.predict(X_test)

print(classification_report(y_test, y_pred_tree))
```

	precision	recall	f1-score	support
0	0.85	0.93	0.89	42
1	0.96	0.90	0.93	72
accuracy			0.91	114
macro avg	0.90	0.92	0.91	114
weighted avg	0.92	0.91	0.91	114

Figure 2: Classification Report Output

5. RESULTS AND DISCUSSION

The Logistic Regression model achieved strong classification performance on the Breast Cancer Wisconsin Dataset. The evaluation metrics indicated that the model successfully classified the majority of tumor samples.

The Confusion Matrix showed only a small number of misclassified cases. High Precision values indicated fewer false positive predictions, while high Recall values demonstrated effective identification of cancer cases. The F1-Score confirmed a good balance between Precision and Recall.

The ROC Curve remained close to the upper-left corner, indicating excellent classification capability. A high AUC Score further demonstrated the model’s effectiveness in separating malignant and benign classes.

When compared with the Decision Tree Classifier, Logistic Regression produced more consistent and reliable results and showed better generalization on unseen data.

6. HANDLING IMBALANCED DATA

Class imbalance occurs when one class contains significantly more samples than another. In classification tasks, this can lead to biased predictions and misleading performance measures.

To address this issue, Logistic Regression was trained using:

```
**class_weight = 'balanced'**
```

This approach automatically assigns higher importance to minority class samples during training.

Benefits include:

- Improved Recall
- Better F1-Score
- Reduced model bias
- More reliable classification performance

The balanced model provided fairer predictions and improved robustness.

7. IMPROVEMENT IDEAS

Although the model achieved good performance, further improvements can be made through:

- Hyperparameter tuning
- Feature selection techniques
- Cross-validation
- Ensemble learning methods

Advanced algorithms such as Random Forest, XGBoost, Gradient Boosting, and Support Vector Machines can also be explored to improve classification accuracy and generalization.

8. CONCLUSION

This project successfully demonstrated the development and evaluation of classification models using the Breast Cancer Wisconsin Dataset. The workflow included data preprocessing, feature scaling, model training, prediction, evaluation, handling class imbalance, and model comparison.

The results showed that Logistic Regression effectively classified breast cancer cases with high accuracy and strong evaluation metric scores. The study also highlighted the importance of using Precision, Recall, F1-Score, ROC Curve, and AUC Score rather than relying solely on accuracy.

Among the evaluated models, Logistic Regression achieved the best overall performance due to its stability, interpretability, and strong predictive capability. The project provided practical experience in applying machine learning techniques to a real-world healthcare classification problem.